



Same Call, Different Camera

A season-long swap-cage audit finds no added safety signal beyond the staff glance

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The problem

A national hardware chain now runs every returned gas bottle past a phone camera before a staff member decides whether it goes back into the swap cage. Little is known about what the camera catches that the clerk’s own glance would have caught anyway.

What we set out to test

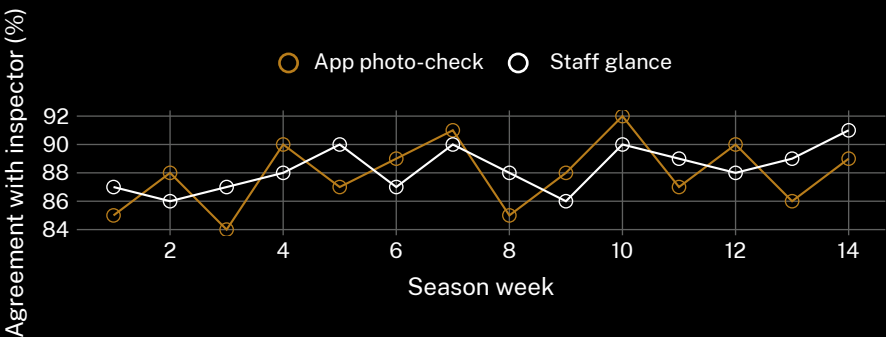
- Do app-graded and staff-graded pass/fail calls differ on the same cylinder?
- Do app rejections catch anything an independent inspector’s blind review confirms and the staff check missed?
- Does the extra photo step change how long a swap takes at the counter?

Study design

- n = 214 cylinders · 6 swap cages · one regional chain · 14-week season
- Each cylinder scored three ways — staff glance, app photo-check, blind inspector reassessment; scoring order rotated weekly
- Counter handling time logged from presentation to refill acceptance

Findings

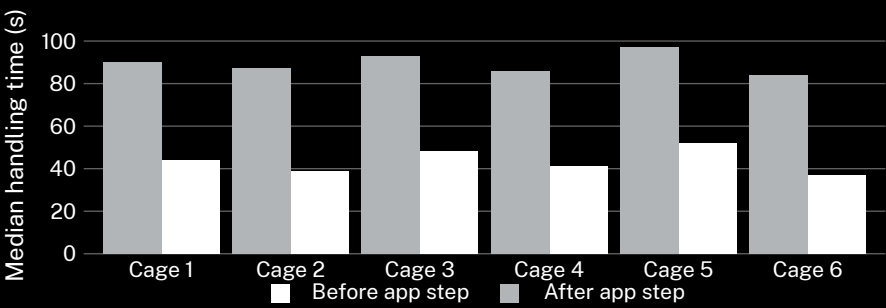
App-graded and staff-graded pass/fail calls agreed with each other on 96% of cylinders. Against the inspector’s blind reassessment, the app matched no better than the staff glance it was added to double-check.



Weekly agreement with the inspector’s blind reassessment stayed within 3 points between the app and the staff check across all 14 weeks (season mean: 88% both).

Implications

The app’s rejections tracked the staff’s, not the inspector’s independent read: nothing it flagged had passed both a clerk and the inspector. Every cage absorbed roughly the same handling-time cost, so the delay reads as a property of the extra step rather than of any one site or crew. Next: trial a staff-only fast lane running alongside the app step for a second season, and re-cost the rollout against that baseline.



Median counter handling time rose from about 40 seconds to about 90 seconds per swap at every cage once the photo-check step joined the counter routine.

Selected references

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We thank the participating swap-cage staff and the chain’s regional safety team. Cylinder identifiers de-identified before analysis; the inspector scored blind to both prior gradings.